I

HEAD INJURIES

or TRAUMATIC BRAIN INJURY (TBI) PREHOSPITAL MANAGEMENT IS AVOIDING SECONDARY BRAIN INJURY BY MAINTAINING NORMOTENSION AND OXYGENATION BY MANAGING CABCDE/MARCH.
ANY LOSS OF CONSCIOUSNESS (LOC) AFTER HEAD TRAUMA = TRANSFER TO HOSPITAL

MAJOR HAEMORRHAGE CONTROL AND ASSESS CONSCIOUS LEVELS (AVPU)

AIRWAY MANAGEMENT (BASED ON AVPU[1])

RESPONDS TO PAIN or O_2 SATS <90%[2]	PROFOUNDLY UNRESPONSIVE or BASIC ADJUNCTS INEFFECTIVE	IMPENDING AIRWAY COLLAPSE or VENTILATORY FAILURE
BASIC AIRWAY ADJUNCTS AND MANOEUVRES (JAW THRUST)	SUPRAGLOTTIC AIRWAY (iGel) CONSIDER PALM SEDATION[3]	A PREHOSPITAL EMERGENCY ANAESTHESIA (PHEA) for DEFINITIVE AIRWAY

CONSIDER C-SPINE IF MECHANISM INIDICATES AND ENVIRONMENT ALLOWS[4]

BREATHING/RESPIRATORY ASSESSMENT

ASSESS BREATHING RATE (<10 VENTILATE), BREATHING DEPTH, BREATHING RHYTHM[5]
MAINTAIN NORMAL OXYGEN SATURATIONS (94-98%)
MAINTAIN LOWER END OF NORMAL END-TIDAL CO_2 (4-4.5kPA or 35-40mmHg)
PERFORM ANY CHEST INTERVENTIONS (BLATOMFC)

IMPACT BRAIN APNOEA[6]

DIRECT HEAD IMPACT CAUSES APNOEA AND CATECHOLAMINE SURGE (↑BP, ↑HR)
COMMENCE VENTILATORY ARREST MANAGEMENT (AIRWAY AND MANUAL VENTILATIONS)
FOLLOW ADVANCED LIFE SUPPORT GUIDELINES IF UNABLE TO DETECT BREATHING

CIRCULATORY ASSESSMENT

ASSESS FOR SHOCK[7][8] MAINTAIN PERIPHERAL PULSES OR SBP >110mmHg.
ASSESS FOR CUSHING'S RESPONSE (HTN, BRADYCARDIA & ABNORMAL RESPS)[9]

GAIN IV/IO ACCESS

GIVE BLOOD PRODUCTS AND TXA 1G FOR POLYTRAUMA HYPOTENSION
GIVE SODIUM CHLORIDE 0.9% FOR HYPOTENSION IN ISOLATED HEAD INJURY[10]

DISABILITY/HEAD ASSESSMENT

FULL GCS[11] NOTING MOTOR SCORE (MOTOR SCORE 4 OR LESS IS SEVERE TBI)
ASSESS PUPILLARY RESPONSE[12] AND BLOOD GLUCOSE (4-10mmol/L)
CONSIDER ANALGESIA IN AGITATED PATIENTS (KETAMINE/IVP)
CONSIDER BENZODIAZEPINES IF PATIENT HAS GENERALISED SEIZURE

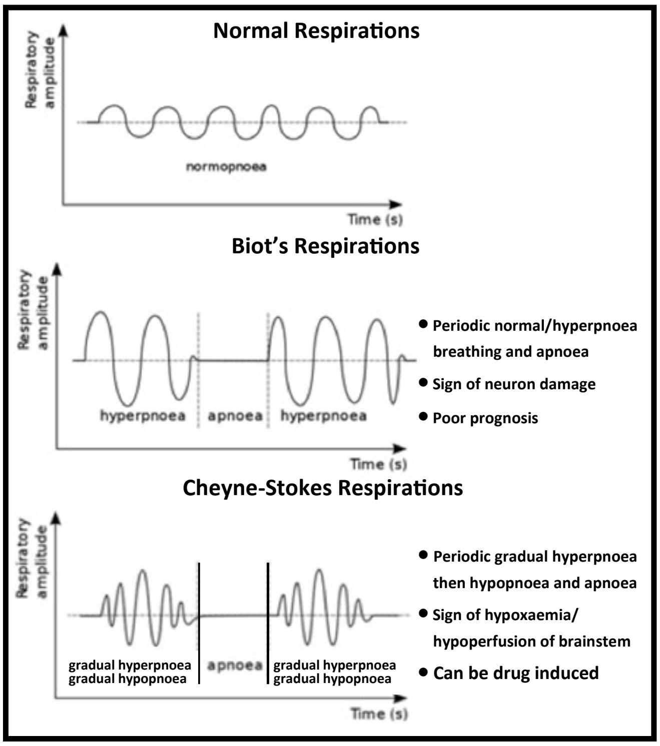
GCS 13-15 MILD TBI (CONCUSSION)	GCS 9-12 MODERATE TBI	GCS <9 / DILATED PUPILS SEVERE TBI

RB HYPERTONIC SALINE 5%[13] WITH GCS <9 AND:
UNILATERAL OR BILATERAL DILATED PUPIL(S) or CUSHING'S RESPONSE

TRANSFER OR MEDEVAC PATIENT 30° HEAD UP TO NEAREST NEURO CAPABILITY/MTC

POINTS TO NOTE

1. **REDUCED RESPONSE LEVELS ARE ASSOCIATED WITH INABILITY TO MAINTAIN AIRWAY.**
2. **DECREASED OXYGEN SATURATIONS COULD BE CAUSED FROM AIRWAY OBSTRUCTION SO SHOULD BE AN INDICATION OF AIRWAY ADJUNCT.**
3. **PHARMACOLOGICALLY ASSISTED LARYNGEAL MASK (PALM) INSERTION GUIDELINES SHOULD ONLY BE USED BY TRAINED PRACTITIONERS (SEE ADVANCED AIRWAY GUIDE). DO NOT DELAY PRIORITY OF GETTING PHEA TRAINED CLINICIAN TO THE PATIENT BEFORE PERFORMING PALM. CONTINUOUS ETCO$_2$ MONITORING IS MANDATORY IN THESE PATIENTS.**
4. **APPROXIMATELY 12% OF SPINAL INJURIES OCCUR WITH SEVERE TBI's AND 6.5% OF CERVICAL SPINAL INJURIES WITH SEVERE TBI's (Pandrich and Demetriades, 2020). CONSIDER FULL IMMOBILISATION (HEAD BLOCKS AND VACUUM MAT).**
5. **BIOTS AND CHEYNE-STOKES BREATHING IS ASSOCIATED WITH SEVERE TBI's (SEE DIAGRAM BELOW). NASAL ETCO$_2$ CAN BE BENEFICIAL IN MONITORING BREATHING RHYTHMS AND SHOULD BE USED IF AVAILABLE.**
6. **IMPACT BRAIN APNOEA OCCURS DURING THE CONCUSSIVE PHASE FROM DIRECT FORCES TO THE HEAD. THE HIGHER LEVEL OF FORCE ON IMPACT TO THE HEAD CORRELATES TO INCREASE TIME OF APNOEA (Spanos, 2021).**
7. **HYPOTENSION IN THE ACUTE PHASE OF TRAUMA SHOULD BE TREATED AS HYPOVOLAEMIA UNTIL PROVEN OTHERWISE.**
8. **PERMISSIVE HYPOTENSION IS NOT TO BE USED IN THE SETTING OF A MODERATE OR SEVERE HEAD INJURY.**
9. **CUSHING'S RESPONSE (IRREGULAR BREATHING, BRADYCARDIA AND HYPERTENSION) INDICATES INTRACRANIAL HERNIATION (CONING) AND BRAIN STEM COMPRESSION.**
10. **MAXIMUM BOLUS OF ISOTONIC CRYSTALLOID FLUID IS 2 LITRES. CONSIDER NEUROGENIC SHOCK IN HEAD INJURIES WHICH MAY NOT RESPOND TO FLUID CHALLENGE AND WILL REQUIRE INOTROPES OR VASOPRESSORS.**
11. **MOTOR SCORE IS THE BEST INDICATION OF SEVERITY OF HEAD INJURY. A MOTOR SCORE OF 4 OR LESS IS CLASSIFIED AS SEVERE TBI.**
12. **UNEQUAL, DILATED OR NON-RESPONSIVE PUPILS IN PATIENTS WITH SIGNIFICANTLY REDUCED CONSCIOUSNESS INDICATES INTRACRANIAL HERNIATION AND NEEDS EMERGENCY CONVEYANCE TO A NEUROSURGICAL CAPABILITY (SEE BELOW)**
13. **HYPERTONIC SALINE 5% AT SLOW INFUSION OF 6ml/kg (UP TO 350ml) OVER 10-20 MINUTES.**
14. **SIGNIFICANT IMPACT TO THE TEMPORAL REGION WITH LOC SHOULD BE CONVEYED AND CONTINUOUSLY MONITORED DUE TO INCREASED RISK OF INTRACRANIAL HAEMORRHAGE. EPIDURAL HAEMATOMAS CAN CAUSE 'TALK AND DIE SYNDROME', WHERE DETERIORATION CAN OCCUR RAPIDLY EVEN IN PATIENTS WITH AN INITIAL GCS 15 ON ASSESSMENT POST LOC.**



Glasgow Coma Scale (GCS)

Score	Eyes	Verbal	Motor
6			Obeys Commands
5		Orientated	Localises Pain
4	Spontaneous Opening	Confused	Withdraws from Pain
3	Open to Voice	Inappropriate Words	Flexion (Decorticate)
2	Open to Pain Response	Incomprehensible Sounds	Extension (Decerebrate)
1	None	None	None

Pupillary Assessment

	Unilateral Dilated Pupil	• Aniscoria (20%) • Head Trauma (CN III Compression) • Direct trauma or medication to eye
	Bilateral Dilated Pupil	• Midbrain injury/lesion (stroke/trauma) • Sympathetic stimulation (drugs, pain) • Oxytocin
	Irregular Pupils	• Direct ocular/orbital trauma
	Dysconjugate Gaze	• Congenital • Frontal Lobe Lesion
	Pinpoint Pupils	• Opioids, haloperidol, nerve agents • Pontine injury/stroke

References

Guideline created 10 May 2023 (V.1.0) in accordance with SFM Clinical Standard Operating Procedure 005 'Management of Head Injured Patients' (2022) and *MERT Standard Operating Procedure 'Traumatic Brain Injury'.*

Lulla, A., et al. (2023) Prehospital Guidelines for the Management of Traumatic Brain Injury - 3rd Edition, *Prehospital Emergency Care*, doi: https://doi.org/10.1080/10903127.2023.2187905

Pandrich, M. J. and Demetriades, A. K. (2020) Prevalence of Concomitant Traumatic Cranio-spinal Injury: A Systematic Review and Meta-analysis, *Neurosurgical Review*, 43(1), pp. 69-77.

Spanos, A. (2021) *Impact Brain Apnoea*. Available at: https://www.impactbrainapnoea.com (Accessed 10 Mar 2023).